EE304 - Communication Lab
Project Report (Component - Cellphone)
Hospital Monitor
Smart Healthcare Monitoring System

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1. Introduction

The **Hospital Monitor** is a communication engineering based smart healthcare monitoring system developed for real-time transmission and monitoring of patient health parameters over a cloud-enabled communication network. The project demonstrates how modern communication technologies such as low-latency data transmission, cloud synchronization, and remote monitoring can be applied in healthcare systems.

The system is designed to simulate a smart hospital environment in which patient vital signs are continuously transmitted from a patient node to a centralized monitoring station. The patient interface acts as the transmitting unit, while the doctor dashboard acts as the receiving and monitoring station. The communication between these nodes is established using Firebase Realtime Database, which behaves as a cloud communication channel.

The project mainly focuses on communication engineering principles including:

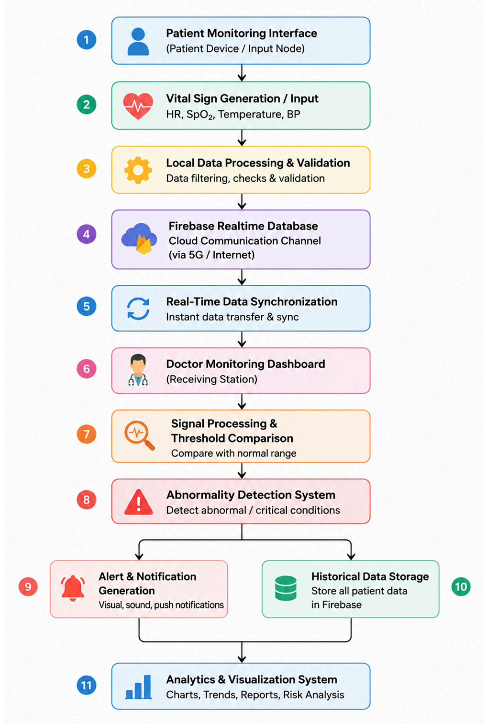
- Real-time digital communication
- Low-latency signal transmission
- Cloud-based communication systems
- Remote healthcare monitoring
- Data synchronization
- Signal propagation and delay analysis
- Emergency communication systems

The application also demonstrates the role of 5G communication in future healthcare systems by enabling faster data transfer, reliable communication links, and continuous patient monitoring. Such systems can be highly useful in smart hospitals, telemedicine, emergency healthcare services, and remote patient monitoring applications.

The major patient parameters monitored in the project include **Heart Rate, Blood Oxygen Level (SpO_2), Body Temperature and Blood Pressure.**

The project provides continuous monitoring, real-time alerts, and historical analysis of patient data, thereby acting as a prototype for modern smart healthcare communication systems.

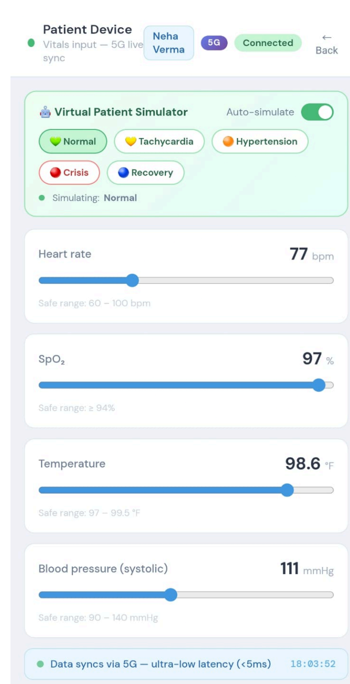
2. Flow Diagram: Overall System Flow



3. METHODOLOGY

The project follows a systematic communication-based methodology for transmitting, monitoring, and analyzing patient health data in real time.

Step 1: Patient Data Acquisition



The patient enters health parameters manually or through a simulation engine. The system generates values for:

- Heart Rate
- SpO_2 Level
- Body Temperature
- Blood Pressure

These parameters behave as biomedical signals in the communication system.

Step 2: Signal Generation and Processing

The generated patient data is processed digitally and prepared for transmission. The simulator creates continuously varying physiological values to imitate real sensor outputs.
Before transmission, a 300ms debounce timer ensures that rapid slider changes are collapsed into a single write, preventing channel flooding.
Different scenarios are simulated such as:

- Normal Condition
- Tachycardia
- Hypertension
- Critical Emergency

- Recovery State

This helps in demonstrating realistic healthcare communication behaviour (just for demonstration purpose)

Step 3: Cloud-Based Data Transmission

The processed signals are transmitted through Firebase Realtime Database, which acts as the cloud communication channel.

The communication system provides:

- Continuous data transfer
- Instant synchronization
- Low communication delay
- Reliable cloud connectivity

The database continuously updates transmitted data so that the receiving station can access it immediately.

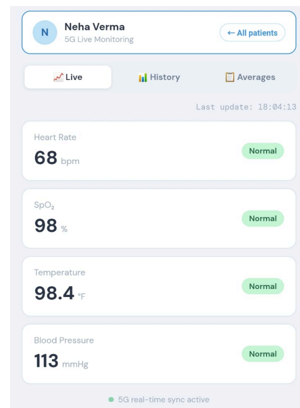
Step 4: Real-Time Monitoring at Receiver Side

The doctor dashboard acts as the **receiving station**. It continuously listens for incoming patient data using real-time synchronization methods.

The receiver performs:

- Signal reception
- Data processing
- Vital parameter visualization
- Real-time monitoring

This demonstrates a practical transmitter–receiver communication model.



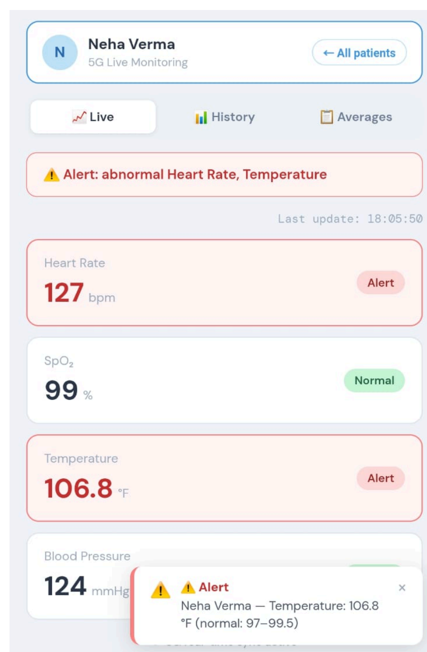
Step 5: Threshold Analysis

After receiving the patient data, the system compares the values against predefined safe ranges.

If the received signal exceeds these limits, the system identifies the condition as abnormal.

Safe Operating Ranges:

Parameter	Safe Range
Heart Rate	60–100 bpm
SpO_2	≥94%
Temperature	97–99.5 °F
Blood Pressure	90–140 mmHg



Step 6: Alert Communication

Whenever abnormal conditions are detected, the communication system generates emergency alerts.

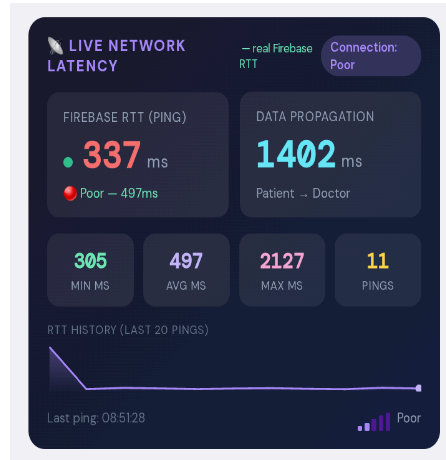
The alert system includes:

- Browser notifications
- Emergency warnings
- Toast messages
- Critical condition indicators

This demonstrates priority-based emergency communication in healthcare systems.

Step 7: Latency Monitoring

The proposed healthcare monitoring system continuously measures communication latency between the patient monitoring interface and the doctor dashboard using Firebase Realtime Database communication.



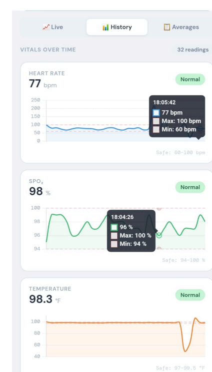
The system evaluates the following communication performance parameters:

- Round Trip Time (RTT)
- Network response time
- Data transmission latency
- Real-time synchronization delay
- Communication stability trends
- Minimum, average, and maximum latency values

These parameters are monitored to analyze the real-time performance, synchronization efficiency, reliability, and responsiveness of the healthcare communication network.

The latency monitoring module also provides graphical visualization of network performance to help identify communication delays and network congestion conditions during real-time patient monitoring.

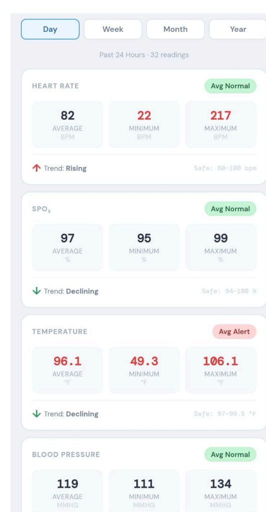
Step 8: Historical Data Storage



All transmitted patient signals are stored in historical records for future analysis and monitoring.
The stored data is used for:

- Trend analysis
- Daily averages
- Weekly averages
- Monthly reports
- Healthcare analytics

Step 9: Data Analytics and Average Calculation



The stored historical patient data is further processed to generate analytical insights and statistical healthcare information.
The analytics module performs:

- Daily average calculation
- Weekly average calculation
- Monthly average calculation
- Trend analysis
- Patient health pattern monitoring
- Graphical data visualization
- Healthcare performance analytics

This analytical processing helps doctors monitor long-term patient conditions, identify abnormal health trends, and improve medical decision-making.

4. IMPLEMENTATION

The implementation of the project mainly focuses on establishing a real-time communication network for healthcare monitoring using modern web and cloud technologies.

Frontend Implementation: The user interface is designed using HTML, CSS3 and JavaScript.
The frontend contains two major modules:
1. Patient Interface: transmitting node
Functions performed:

- Input of health parameters
- Simulation of patient conditions
- Transmission of vital signals
- Continuous data updates

2. **Doctor Dashboard: receiving and monitoring station**
Functions performed:

- Real-time signal reception
- Patient monitoring
- Alert generation
- Visualization of health parameters
- Communication delay monitoring

The dashboard continuously updates without requiring page refresh, thereby enabling real-time communication.
3. Risk Score Module
A triage risk scoring algorithm computes a 0–100 score for each patient based on how far each vital deviates from its safe range. Scores map to Low (0–19), Moderate (20–49), and High (50+) risk badges on the patient list.
4. Demo Patients Module
Three preconfigured demo patients (Normal, ICU, Critical) push simulated data to Firebase every 800ms from the doctor side, enabling multi-patient monitoring demonstration without a second physical device.

Backend Implementation: Firebase Realtime Database is used as the cloud communication infrastructure.
Functions of Firebase:

- Cloud data storage
- Real-time synchronization
- Continuous communication
- Instantaneous data synchronization
- Signal transfer between nodes

The database acts as the communication medium connecting the transmitting and receiving stations.

Real-Time Synchronization: The project uses Firebase listeners to continuously monitor changes in patient data. Example:

```
db.ref('patients').on('value', snap => {allPatientsData = snap.val() || {}; });
```

This allows instantaneous signal reception and synchronization between users.

Latency Monitoring Implementation: The project implements communication latency monitoring using round-trip time (RTT) calculations.The system monitors:

- Network response time
- Communication latency
- Real-time synchronization delay
- Minimum, average, and maximum latency values
- Communication stability trends

Latency values are updated continuously to analyze network performance and communication reliability.

Notification System Implementation: The notification subsystem is implemented using browser notification APIs. The system generates alerts when:

- Heart rate exceeds limits
- Oxygen level decreases
- Temperature becomes abnormal
- Blood pressure crosses thresholds

This demonstrates emergency communication handling.
Alert fatigue prevention - The system only fires alerts on new threshold crossings, not on every update. A state tracker (_prevAlertKeys) records which vitals were in alert in the previous cycle, so sustained abnormal readings do not repeatedly trigger notifications.

5. OPERATION

The operation of the project follows the working principle of a real-time digital communication system.
Step 1: System Initialization
When the application starts, the user selects either: Patient Mode OR Doctor/Nurse Mode
The communication network is then initialized.

Step 2: Patient Signal Transmission
The patient enters or simulates vital parameters.
These signals are transmitted continuously to the cloud communication channel.

Step 3: Cloud Synchronization
The Firebase communication server synchronizes the data instantly between the transmitting and receiving stations.
This ensures:

- Real-time updates
- Low-latency communication
- Reliable data transfer

Step 4: Receiver Monitoring
The doctor dashboard continuously receives and displays incoming patient signals.
The receiving station performs:

- Signal visualization
- Threshold comparison
- Risk analysis
- Patient monitoring

Step 5: Emergency Alert Operation
If abnormal signals are received:

- Alerts are generated immediately
- Warning notifications are displayed
- Critical conditions are highlighted
- Emergency communication is initiated

This enables quick medical response.

Step 6: Data Logging and Analytics
All communication records are stored in the historical database.
The stored information is used for:

- Patient history tracking
- Trend analysis
- Statistical calculations
- Long-term monitoring

Step 7: Continuous Communication Cycle
The entire process continues continuously in real time, thereby creating a complete smart healthcare communication system capable of remote patient monitoring and emergency healthcare communication.
Every 10 seconds, a historical snapshot is pushed to the `patientHistory/` node in Firebase, building a time-ordered log used for trend analysis.

6. Limitations

- Uses simulated signals instead of real sensors.
- Depends on internet connectivity.
- No dedicated hardware communication module.
- No encryption or secure channel implementation.
- Limited to software-based simulation.

7. Future Enhancements

Future improvements may include:

• Integration with biomedical sensors
• Dedicated wireless communication modules
• AI-assisted healthcare analytics
• ECG waveform transmission
• Secure encrypted communication
• IoT-based healthcare nodes
• Mobile communication applications

8. Conclusion

The “Hospital Monitor” successfully demonstrates the application of communication engineering principles in healthcare monitoring systems.

The project highlights:

- Real-time digital communication
- Low-latency healthcare networks
- Remote monitoring systems
- Cloud-based communication channels
- Emergency alert transmission
- Telemedicine infrastructure

The project acts as a prototype for future smart healthcare systems based on advanced wireless communication technologies such as 5G and IoT.